

Appendix

Fast, Faithful, or Diverse — Pick Two: Decoding Strategy Trade-offs in Open-Ended Text Generation with GPT-2

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Repository: <https://github.com/affifamca/Decoding-Strategy-Tradeoffs>

1. References

1.1 Academic literature

- Fan, A., Lewis, M., & Dauphin, Y. (2018). Hierarchical Neural Story Generation. Proceedings of ACL 2018, 889–898. — top-k sampling, the family sample_t0.7/1.0/1.2 belong to.
- Hermann, K. M., Kočiský, T., Grefenstette, E., Espeholt, L., Kay, W., Suleyman, M., & Blunsom, P. (2015). Teaching Machines to Read and Comprehend. Advances in Neural Information Processing Systems (NeurIPS) 28. — origin of the CNN/DailyMail corpus.
- Holtzman, A., Buys, J., Du, L., Forbes, M., & Choi, Y. (2020). The Curious Case of Neural Text Degeneration. Proceedings of ICLR 2020. — introduces nucleus (top-p) sampling; directly motivates the sample_t*_topp* configurations and the repetition-rate metric.
- Lin, C.-Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. Text Summarization Branches Out: Proceedings of the ACL 2004 Workshop, 74–81.
- Meister, C., Cotterell, R., & Vieira, T. (2020). If Beam Search Is the Answer, What Was the Question? Proceedings of EMNLP 2020, 2173–2185. — critiques the assumption that more search (higher beam width) monotonically improves output quality; directly relevant to the H1 finding that greedy outperforms beam here.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI Technical Report. — GPT-2, the model under test.
- See, A., Liu, P. J., & Manning, C. D. (2017). Get To The Point: Summarization with Pointer-Generator Networks. Proceedings of ACL 2017, 1073–1083. — establishes the non-anonymized CNN/DailyMail summarization setup (dataset version 3.0.0) used here.
- Welleck, S., Kulikov, I., Roller, S., Dinan, E., Cho, K., & Weston, J. (2020). Neural Text Generation with Unlikelihood Training. Proceedings of ICLR 2020. — motivates treating repetition rate as a quality signal distinct from ROUGE.

1.2 Datasets, models and software

Non-academic sources are cited here only in the categories the guidelines permit: dataset documentation, model cards and official software documentation.

Hugging Face. gpt2 model card. <https://huggingface.co/gpt2>

Hugging Face. cnn_dailymail dataset card. https://huggingface.co/datasets/cnn_dailymail — underlying data is Hermann et al. (2015) and See et al. (2017).

Lhoest, Q. et al. (2021). Datasets: A Community Library for Natural Language Processing. Proceedings of EMNLP 2021: System Demonstrations, 175–184. — [huggingface/datasets](https://huggingface.co/datasets), used to load and cache the corpus.

Wolf, T. et al. (2020). Transformers: State-of-the-Art Natural Language Processing. Proceedings of EMNLP 2020: System Demonstrations, 38–45.

Google Research. rouge-score Python package documentation. <https://github.com/google-research/google-research/tree/master/rouge> — reference ROUGE implementation used for all quality scores.

Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

2. Reproducibility

Everything here can be reproduced from the repository linked above. One GPT-2 model and one cached CNN/DailyMail subset feed all three experiments; only the decoding configuration passed to `model.generate()` changes between runs.

Step	Command	Approx. runtime (CPU)
Verify wiring	<code>python -m src.prepare --smoke</code>	~1 min
H1 · Quality	<code>python -m src.exp1_quality</code>	~35 min
H2 · Diversity	<code>python -m src.exp2_diversity</code>	~73 min
H3 · Cost	<code>python -m src.exp3_cost</code>	~10 min
Figures	<code>python -m src.figures</code>	< 1 min

Seeds are fixed (`config.SEED = 42`; per-sample seeds `SEED..SEED+4` for the 5 resamples in H2) and recorded per row in the result CSVs, so every number on the poster can be traced back to a committed CSV under `results/`.

Full decoding-configuration grid

Config	Specification
Greedy	<code>do_sample=False, num_beams=1</code>
Beam (k=4)	<code>do_sample=False, num_beams=4, early_stopping=True</code>
Sample T=0.7	<code>do_sample=True, temperature=0.7</code>
Sample T=1.0, p=0.9	<code>do_sample=True, temperature=1.0, top_p=0.9</code>
Sample T=1.2, p=0.95	<code>do_sample=True, temperature=1.2, top_p=0.95</code>

Model held constant across all configs: gpt2 (124M, frozen, no fine-tuning), `max_new_tokens = 60` (summarisation) or 40 (continuation). Perplexity is computed as the model's own loss on its generated text, so it is only a fluency proxy relative to GPT-2's own distribution, not an external quality measure.

3. Use of Generative AI

Claude (Anthropic) was used for research and study purposes only.